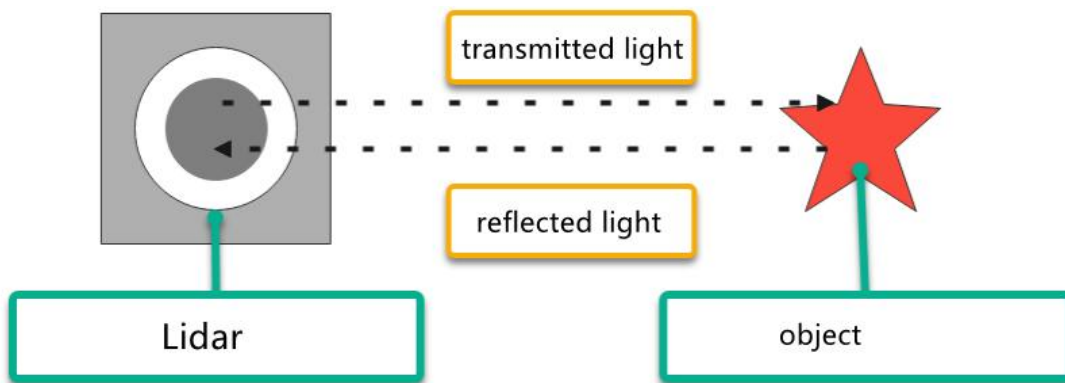


Lidar Working and Ranging Principle

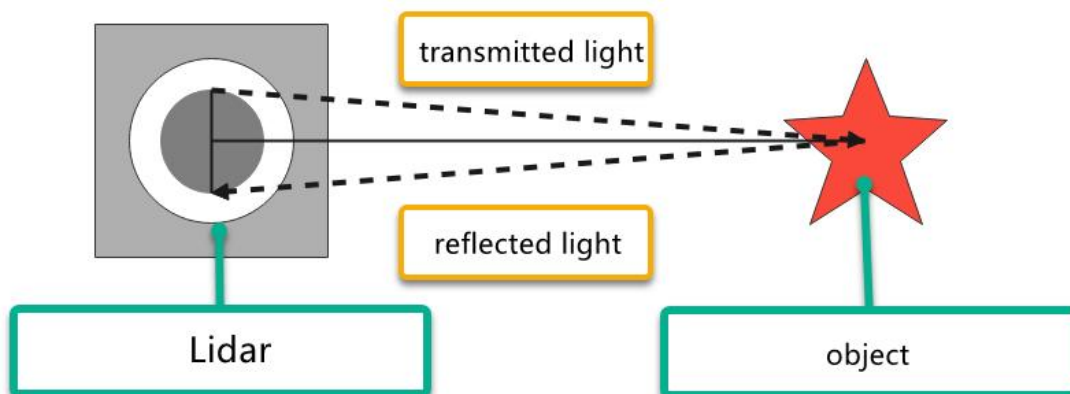
1. Lidar Ranging

Two common methods are employed by Lidar to determine the distance to a target: triangulation and Time of Flight (TOF).

In the case of TOF, as depicted in the diagram, the Lidar initially projects light onto the object. The object reflects the light directly back to the Lidar, which calculates the time it takes for the light to return. The distance between the object and the Lidar is then obtained by multiplying this time by the speed of light.



In triangulation, as illustrated in the diagram, the Lidar undergoes adjustments during manufacturing to ensure that the light does not directly strike the object. Instead, it is projected at a specific angle, a pre-set value that remains constant during operation. The distance from the object to the Lidar can be calculated by incorporating this angle into trigonometric functions.



2. Lidar Working Result

Its working result is as follows. Lidar will emit light and shine it on the object surface. When receiving the light reflected by the object, Lidar will mark the contour of the object at the position where the light is reflected.

